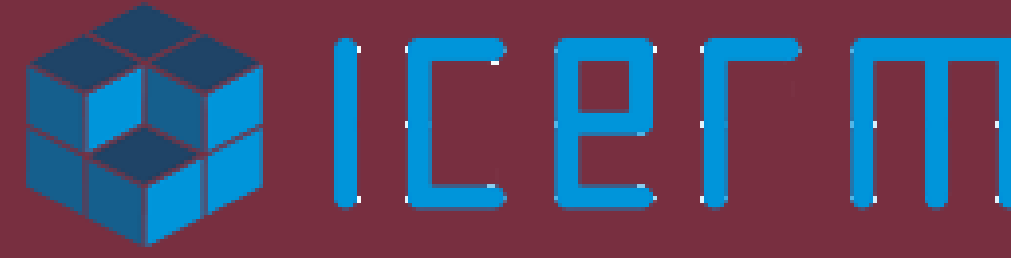




Generalized EXTRA SGLD: Eliminating Network Bias in Decentralized Bayesian Learning

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Introduction

Assume there are N agents (*e.g.*, mobile devices, hospitals, or data centers) connected through a network $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ with $\mathcal{V} = 1, 2, \dots, N$ representing agents and $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ being the set of connections.

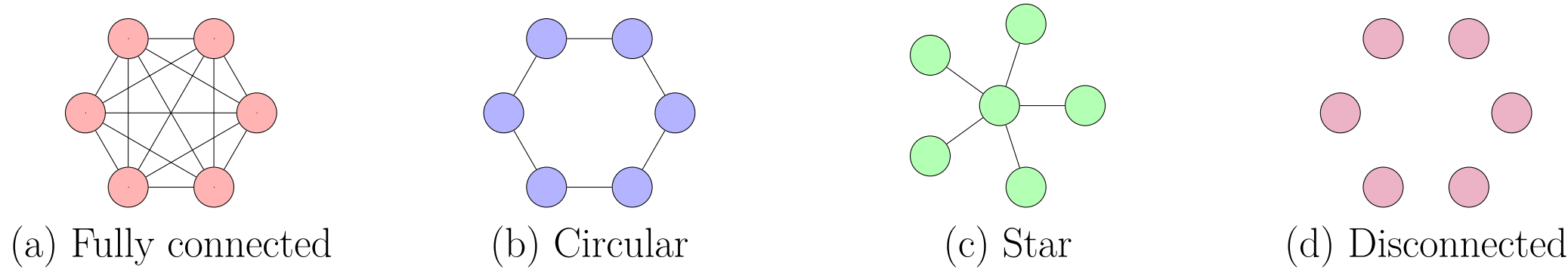


Figure 1: Different types of network structures

Global Dataset: $Z = \cup_{i=1}^N Z_i$ (disjoint subsets across N agents). Local Data: Agent i only has access to its local subset Z_i . Likelihood: Due to i.i.d. data, the log-likelihood is additive. The **objective** is to sample from the global posterior $\pi(x) \propto e^{-f(x)}$ where,

$$f(x) = \sum_{i=1}^N f_i(x), \text{ and } f_i(x) := -\sum_{z \in Z_i} \log p(z|x) - \frac{1}{N} \log p(x).$$

Each agent i manages a local objective $f_i(x)$ defined by its own data and a portion of the prior.

The Challenges: Agents must collaboratively sample from the global density $\pi(x)$ through local communication only, without ever sharing their private Z_i .

The Problem: Network-Induced Bias

Standard Decentralized SGLD (DE-SGLD)¹ allows agents to sample from the posterior by performing three simultaneous steps at each iteration k :

$$x_i^{(k+1)} = \sum_{j \in \Omega_i} W_{ij} x_j^{(k)} - \eta \tilde{\nabla} f_i(x_i^{(k)}) + \sqrt{2\eta} w_i^{(k+1)}, \quad (1)$$

where agents average their local variables x_i with immediate neighbors using a doubly stochastic weight matrix W , and $\tilde{\nabla} f_i$ is an unbiased estimate of the full gradient $\nabla f = \tilde{\nabla} f + \xi$ to handle large datasets ($n_i \gg 0$).

Why is DE-SGLD Inexact? As $k \rightarrow \infty$ and $x^\infty = \mathbf{x}_*$, then

$$\mathcal{W}x^\infty = x^\infty, \quad x^\infty = x^\infty - \eta \nabla f(x^\infty) \quad (2)$$

which means $\nabla f_i(x^*) = 0$ for every agent i . In decentralized settings, local datasets are different, so their individual optimizers are different. Therefore, no such x^* exists, and thus **Network-Induced Bias** is introduced.

Contributions

- Inspired by the EXTRA³ method, introduce the *Generalized EXTRA SGLD*² method to eliminate network-induced bias, ensuring convergence to the exact global posterior even in the full-batch setting.
- Provide the first non-asymptotic proof that the algorithm converges **geometrically fast (linearly)** to the target distribution in 2-Wasserstein distance.
- Demonstrate that the method outperforms standard DE-SGLD by a factor of $\mathbf{L}$ (smoothness coefficient) and achieves superior sampling accuracy across various network topologies.

Generalized EXTRA SGLD

- **Assumption 1:** Let $\mathcal{S}_{\mu,L} : \mathbb{R}^d \mapsto \mathbb{R}$ denote the functions that are μ -strongly convex and L -smooth, *i.e.*, for any $g \in \mathcal{S}_{\mu,L}(\mathbb{R}^d)$, $\frac{L}{2} \|x - y\|^2 \geq g(x) - g(y) - \nabla g(y)^\top (x - y) \geq \frac{\mu}{2} \|x - y\|^2$, $\forall x, y \in \mathbb{R}^d$. We assume the component functions $f_i \in \mathcal{S}_{\mu,L}(\mathbb{R}^d) \quad \forall i = 1, 2, \dots, N$.
- **Assumption 2:** For every $i = 1, 2, \dots, N$ and $k = 0, 1, 2, \dots$, the gradient noise has the property that $\mathbb{E}[\xi_i^{(k+1)} | \mathcal{F}_k] = 0$ and $\mathbb{E} \|\xi_i^{(k+1)}\|^2 \leq \sigma^2$, where $\mathcal{F}_k$ is the natural filtration up to (and including) time k .
- **Assumption 3:** Consider another doubly-stochastic matrix $\tilde{W}$ such that (i) $\text{null}\{W - \tilde{W}\} = \text{span}\{\mathbf{1}_N\}$, $\text{null}\{I_N - \tilde{W}\} \supseteq \text{span}\{\mathbf{1}_N\}$, and (ii) $\tilde{W} \succ 0$, $\frac{I_N + \tilde{W}}{2} \succ \tilde{W} \succ W$.

Then, the **EXTRA SGLD** updates for N agents are given as,

$$x^{(k+2)} = \mathcal{W}x^{(k+1)} - \eta \nabla F(x^{(k+1)}) - \eta \xi^{(k+1)} + \sqrt{2\eta} w^{(k+2)}, \quad (3)$$

$$x^{(k+1)} = \tilde{\mathcal{W}}x^{(k)} - \eta \nabla F(x^{(k)}) - \eta \xi^{(k)} + \sqrt{2\eta} w^{(k+1)}, \quad (4)$$

where $\mathcal{W} = W \otimes I_d$, $\tilde{\mathcal{W}} = \tilde{W} \otimes I_d$, $x^{(k)} = \left[(x_1^{(k)})^\top, \dots, (x_N^{(k)})^\top \right]^\top \in \mathbb{R}^{Nd}$ and

$$w^{(k)} = \left[(w_1^{(k)})^\top, \dots, (w_N^{(k)})^\top \right]^\top, \quad k = 0, 1, 2, \dots,$$

and we assume that the mixing matrices $W, \tilde{W}$ satisfy the assumptions above.

The **generalized EXTRA SGLD** updates for N agents are given as follows

$$x^{(k+1)} = \tilde{\mathcal{W}}x^{(k)} - \eta (\nabla F(x^{(k)}) + v^{(k)}) - \eta \xi^{(k)} + \sqrt{2\eta} w^{(k+1)} \quad (5)$$

$$v^{(k+1)} = v^{(k)} - \mathcal{U}(v^{(k)} + \nabla F(x^{(k)}) - \mathcal{B}x^{(k)}) - \mathcal{U}\xi^{(k)} + \mathcal{U}\left[\frac{2}{\eta}w^{(k+1)}\right], \quad (6)$$

where the matrix $\mathcal{U} = \tilde{\mathcal{W}} - \mathcal{W}$, and the matrix $\mathcal{B} = B \otimes I_d$ has the property $\mathbf{1}_N^\top B = c$ with $c \in \mathbb{R}$.

Convergence Analysis

Under the assumptions and suitable network parameters (η, h) , the **average iterate** satisfies,

$$\mathcal{W}_2(\mathcal{L}(\bar{x}^{(K)}), \pi) \leq \mathcal{O}\left(\frac{L\rho^K}{\sqrt{N}}\right) + \mathcal{O}\left((1 - \mu\eta)^K\right) + \mathcal{O}(\sqrt{\eta}\Xi)$$

and the **agent iterate** satisfies,

$$\frac{1}{N} \sum_{i=1}^N \mathcal{W}_2(\mathcal{L}(x_i^{(K)}), \pi) \leq \mathcal{O}\left(\frac{\eta}{\sqrt{N}}\right) + \mathcal{O}(\sqrt{\eta}\Xi) + \mathcal{O}\left(\frac{\rho^K}{\sqrt{N}}\right) + \mathcal{O}\left((1 - \mu\eta)^K\right).$$

where $\rho \in (0, 1)$ is a constant defined in terms of the network properties (*i.e.*, spectral gap) and hyperparameters of the algorithm, $\mathcal{O}(\cdot)$ hides the algorithmic constants, and $\mathcal{W}_2$ represents 2-Wasserstein distance to compare two probability distributions.

Comparison

For DE-SGLD, under the assumptions in Theorem 1 in [1], as $\varepsilon \rightarrow 0$,

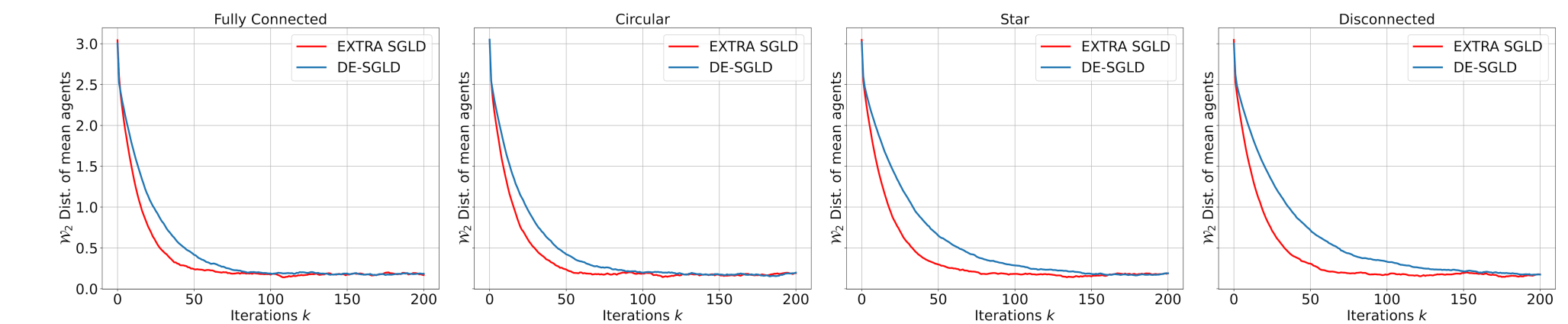
$$\frac{1}{N} \sum_{i=1}^N \mathcal{W}_2^{\text{de-sgld}}(\mathcal{L}(x_i^{(K)}), \pi) \leq \mathcal{O}(\varepsilon), \text{ when } K \geq K^{\text{de-sgld}} = \mathcal{O}\left(\frac{L^4 d}{\varepsilon^2 \mu^3}\right)$$

For EXTRA SGLD, under the assumptions in Theorem 4 in [2], as $\varepsilon \rightarrow 0$,

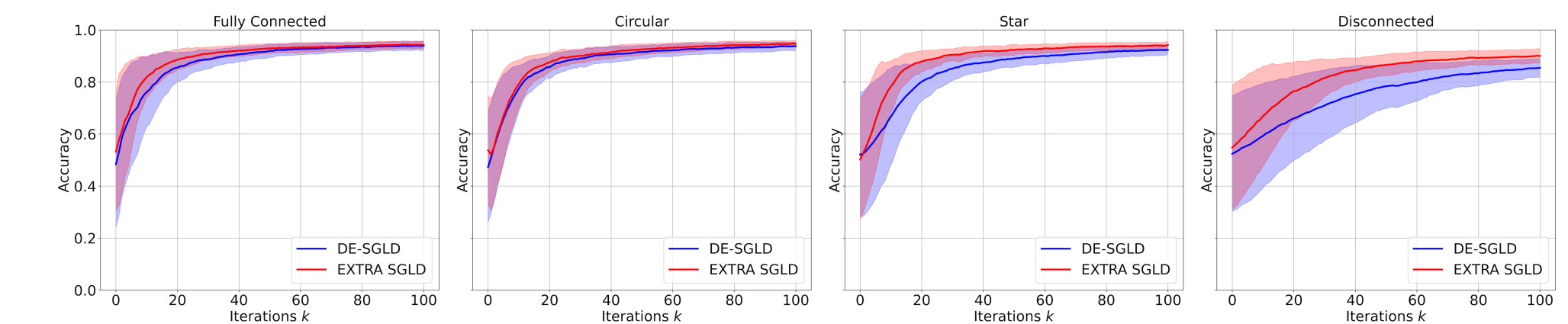
$$\frac{1}{N} \sum_{i=1}^N \mathcal{W}_2^{\text{extra-sgld}}(\mathcal{L}(x_i^{(K)}), \pi) \leq \mathcal{O}(\varepsilon), \text{ when } K \geq K^{\text{extra-sgld}} = \mathcal{O}\left(\frac{L^2 d}{\varepsilon^2 \mu^3}\right)$$

Numerical Experiments

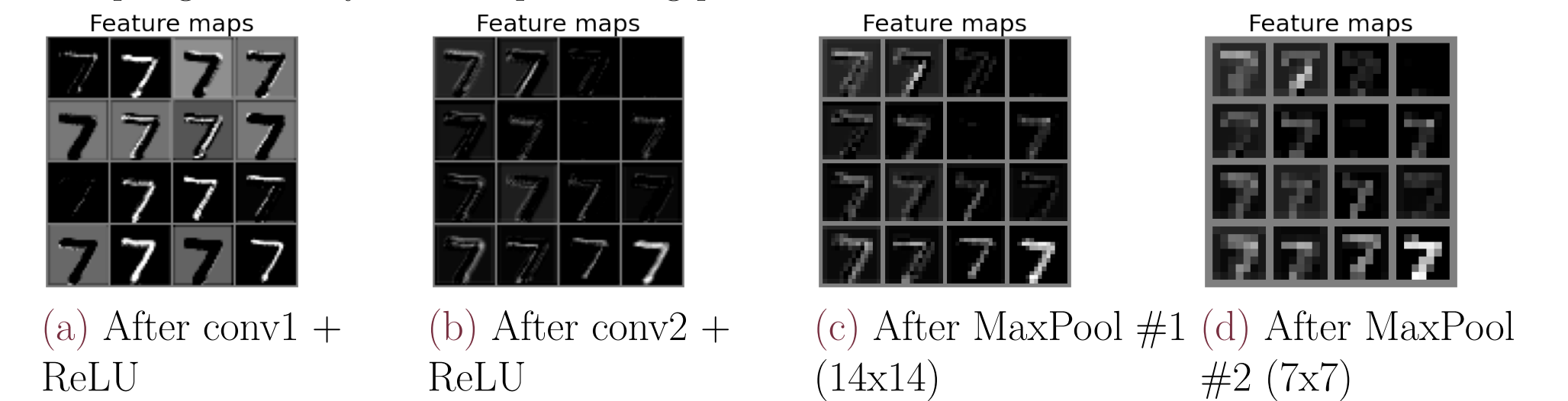
- Sampling for a Bayesian linear regression problem with synthetic data:



- Sampling for a Bayesian logistic regression problem with real data (UCI Breast Cancer Data):



- Sampling for a Bayesian deep learning problem with MNIST data:



	FCN	CN	SN	FDN (Consensus)	FDN (Ensemble)
DE-SGLD	98.80%	98.79%	99.08%	11.79%	99.08%
EXTRA SGLD	98.91%	98.92%	98.96%	15.31%	99.18%

Table 1: MNIST classification accuracies across different network topologies

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